

FineSat: Enhancing GNSS Signals for High-precision Sensing

Anlan Yu[†], Xuanzhi Wang[†], Pei Wang[‡], Jinkun Li[§], Xujun Ma[‡], Zhiqing Hong[¶],
Haotian Wang[§], Yi Ding[◊], and Daqing Zhang^{†‡}

[†]School of Computer Science, Peking University, China [‡]SAMOVAR, Telecom SudParis, IP Paris, France

[§]JD Logistics, China [¶]Rutgers University, USA [◊]University of Texas at Dallas, USA

yal6040@pku.edu.cn, xuanzhiwang@stu.pku.edu.cn, pei.wang@telecom-sudparis.eu,

lijinkun14@jd.com, xujun.ma@telecom-sudparis.eu, zhiqing.hong@rutgers.edu,

wanghaotian18@jd.com, yi.ding@utdallas.edu, dqzhang@sei.pku.edu.cn

Abstract—Wireless sensing technologies have shown significant promise in various applications, but their spatial coverage is confined to the vicinity of the transmitters, limiting their applicability in broader environments. In this paper, we introduce an innovative wireless sensing approach based on the globally covered Global Navigation Satellite System (GNSS) signals. While GNSS signals have been widely used in remote sensing to monitor slow changes in the Earth's surface, like sea level and snow depth, their ability to accurately detect highly dynamic target motions, such as human respiration, gestures, and intrusions, remains unclear. The main challenge arises from the interference brought by the large-scale satellite movement and severe GNSS signal errors. In this study, we present a novel GNSS signal enhancement system named FineSat to address these interference. Specifically, we first utilize polynomial representations to cancel satellite movement interference. Then, based on the analysis of GNSS signal errors, we propose a signal differential processing module to mitigate the errors. We implement our system on commercial devices and validate its performance in three sensing applications: respiration monitoring, gesture recognition, and intrusion detection. Results show that we achieve 0.42 bpm mean absolute error in respiration monitoring, 96.5% average accuracy in gesture recognition, and 98.6% accuracy in intrusion detection.

I. INTRODUCTION

Over the past decade, wireless sensing has achieved significant advancements due to its unique advantages, such as being non-intrusive and privacy-preserving. The fundamental principle of wireless sensing is that the motion of a target affects signal propagation, resulting in signal variations that can be analyzed to infer the target's motion. Various wireless signals, including millimeter waves [1]–[3], acoustics [4]–[6], ultra-wideband signals [7], [8], Wi-Fi [9]–[13], LTE [14], and RFID [15] have been utilized to develop a wide range of sensing applications such as indoor tracking, human activity recognition, and vital sign monitoring.

However, the effective sensing area is constrained by the limited signal coverage. The received signal attenuates as the distance between the transmitter and receiver increases. Beyond a certain threshold, the signal becomes nearly undetectable, making target motion extraction unachievable. For example, the effective communication range of Wi-Fi signals typically extends up to 500 meters around the transmitter without obstruction [16]. Although cellular signals, such as LTE and 5G, can cover distances up to 5 kilometers from the base station, many regions still experience poor signal coverage due to insufficient base station deployment [17].

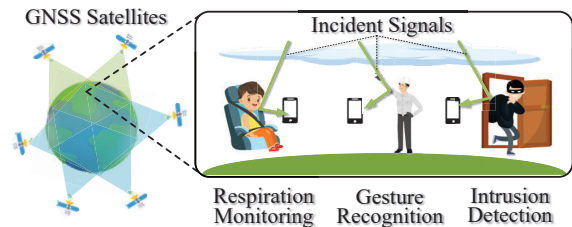


Fig. 1: High-precision sensing of human activities with ubiquitous GNSS signals.

Alternatively, satellite-based navigation signals (i.e., Global Navigation Satellite System, GNSS) offer promising potential with extended signal coverage for sensing. Unlike fixed ground-based stations or access points, more than 100 navigation satellites orbit the Earth at altitudes of 20,000 to 36,000 kilometers, continuously broadcasting positioning signals that provide global and freely accessible coverage [18]. Similar to Wi-Fi or LTE signals, GNSS signal propagation can be affected by target motion, leading to signal variations [19]. By analyzing these variations, it becomes possible to develop novel GNSS sensing applications.

In the field of remote sensing, prior studies employ GNSS reflection signals to monitor slow changes in Earth's surface environment, including sea level, snow depth, and soil moisture [20]–[23]. The underlying principle is that changes in satellite elevation *slowly* alter multi-path signals, leading to *gradual long-term variations* in the received signal. By analyzing the amplitude and frequency of these variations, Earth's surface environment can be monitored. As satellite elevation changes slowly, it is necessary to take a prolonged observation period lasting from a few minutes to several hours. In contrast, highly dynamic target motions, such as human respiration, gestures, and intrusions, *instantly* alter the multi-path signals, causing *abrupt short-term variations* in the received signal. As a result, traditional remote sensing approaches, which focus on analyzing gradual, long-term signal variations, are inadequate for sensing highly dynamic target motions, limiting the broader applicability of GNSS sensing technologies.

To achieve high-precision sensing of highly dynamic target motions and broaden the application of GNSS sensing technologies, we meet two primary challenges as follows:

- **Interference of large-scale satellite movement:** GNSS

satellites orbit the Earth at speeds of up to 4 kilometers per second, causing signal path lengths to change by several hundred meters per second. In contrast, target motions, such as human respiration or gestures, occur at much slower speeds, resulting in signal path length changes of merely a few centimeters per second. As a result, the large-scale movement of satellites introduces substantial variations that serve as interference, overshadowing the subtle signal variations generated by target motions.

- **Interference of severe GNSS signal errors:** After the long-distance signal propagation from the satellite to the receiver, the received signal strength decreases to about -140 dBm and suffers from three sources of signal errors, including satellite-related errors (e.g., satellite clock errors), receiver-related errors (e.g., receiver clock errors) and atmospheric errors (e.g., ionosphere and troposphere errors). These errors can lead to positioning inaccuracies of up to 30 meters [24]. In sensing applications, such errors introduce substantial interference with the subtle signal variations caused by target motions at the sub-centimeter level.

To address these challenges, we present a novel GNSS signal enhancement system named *FineSat*, that systematically **models** and **cancels** interference caused by satellite movement and signal errors in the context of target motion sensing. For the first challenge, we find that the satellite orbit can be accurately approximated using a polynomial function over short-term observations. Consequently, the interference caused by the large-scale satellite movement can be estimated through polynomial fitting. By subtracting this estimated satellite movement component from the signal variations, our system isolates the target motion component. For the second challenge, we perform an in-depth analysis of the three primary sources of GNSS errors within our modeling, including satellite-related errors, receiver-related errors and atmospheric errors. Through comparative experiment, we identify receiver-related errors as the main obstacle to achieving high-precision target motion sensing. As these errors are consistent across various satellites, our system incorporates a module that mitigates them by applying signal differentials across satellites.

The *FineSat* system enhances GNSS signals, thereby enabling high-precision sensing of highly dynamic target motions. We evaluate *FineSat* across three representative high-precision sensing applications—respiration monitoring, gesture recognition, and intrusion detection—to validate its efficacy and showcase the potential of GNSS signals in various sensing tasks. The widespread coverage of GNSS signals allows these applications to function in more diverse environments, addressing the coverage limitations encountered with other sensing signals.

In summary, our contributions are as follows:

- A comprehensive GNSS sensing model is introduced for the first time to analyze diverse interferences affecting the sensing of highly dynamic target motions, including large-scale satellite movements and severe GNSS signal errors.
- Based on the model, we present a novel GNSS signal

enhancement system named *FineSat* to cancel the interference caused by large-scale satellite movements using polynomial fitting and removes severe GNSS errors through signal differential across satellites.

- We implement *FineSat* on commercial equipment and evaluate the system on three typical sensing applications: respiration monitoring, gesture recognition, and intrusion detection. Experiment results show that our system achieves outstanding performance across various environments.

II. PRELIMINARY OF GNSS

Before presenting our design, we introduce the background of the Global Navigation Satellite System and satellite signals.

A. The GNSS System and Satellite Signals

Global Navigation Satellite Systems (GNSS), including GPS, BeiDou, and GLONASS, provide worldwide positioning and navigation services. GNSS satellites orbit at altitudes of 20,000 to 23,000 kilometers with a linear velocity of approximately 3,900 m/s and an angular velocity of 1.5×10^{-4} rad/s [25]. As these signals traverse the atmosphere, they weaken, reaching the Earth's surface with strengths typically between -130 dBm and -160 dBm [26]. Earth-based GNSS receivers can usually connect with 20 to 30 satellites simultaneously [27], receiving signals across frequency bands such as GPS's L1 (1575.42 MHz), L2 (1227.60 MHz), and L5 (1176.45 MHz) [25].

B. GNSS Raw Measurements in Commodity Receivers

Commodity GNSS receivers acquire raw GNSS measurement data from satellites for localization and navigation. The key measurements include carrier-to-noise ratio, pseudorange and carrier phase.

Carrier-to-noise ratio (C/N0) measures the signal power relative to noise power density. In most commonly used protocols like NMEA 0183 [28] and RTCM [29], C/N0 is rounded to the nearest integer, *limiting the granularity needed for high-precision applications such as respiration monitoring.*

Pseudorange is the apparent distance between the satellite and receiver derived from the ranging code. Commercial devices offer only meter-level accuracy due to the limited precision of civil ranging code [30], *making it inadequate for applications demanding high resolution.*

Carrier Phase provides precise information about signal propagation distance by measuring the phase difference between received and transmitted carrier waves:

$$\Phi(t) = \varphi_r(t) - \varphi_s(t) = \hat{\varphi}_r(t) - \varphi_s(t) + 2\pi N, \quad (1)$$

where $\varphi_r(t)$ is the actual carrier phase at the receiver, $\hat{\varphi}_r(t)$ is the reported carrier phase, N is the unknown integer ambiguity due to the periodicity of the carrier phase, and $\varphi_s(t)$ is the transmitted carrier phase. *Due to hardware design, carrier phase achieves higher precision than pseudorange, enabling centimeter- or even millimeter-level accuracy in positioning, thus offering significant potential for high-precision sensing.*

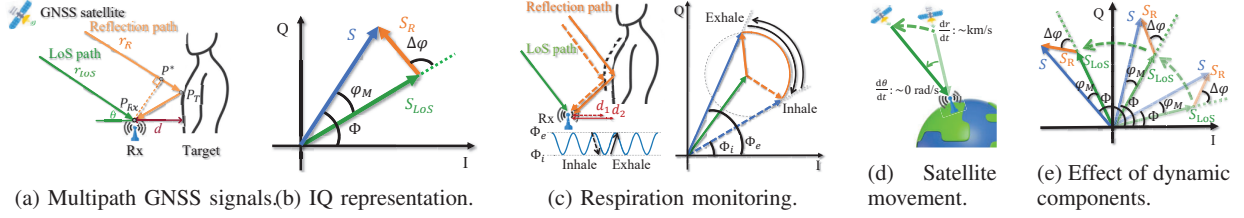


Fig. 2: Modeling the dynamic components of sensing signals.

III. GNSS SENSING AND INTERFERENCE MODELING

In this section, we present a GNSS sensing model to examine interferences on target motion sensing. We first model the dynamic components of sensing signals to analyze interference of satellite movement, followed by an analysis of interference of various GNSS signal errors.

A. Modeling the Interference of Satellite Movement on Sensing

GNSS sensing signals consist of two dynamic components: target motion and satellite movement. Target motion, such as human respiration, occurs at velocities of merely a few millimeters per second, whereas satellite movement can reach 4 km/s. Consequently, satellite movement component tends to overshadow target motion. We begin by modeling the target motion component in the absence of satellite movement, followed by modeling how interference of satellite movement component obscures target motion.

The target motion component originates from the multipath effect of GNSS signals. The signals travel from the satellite to the receiver via multiple paths, including both Line-of-Sight (LoS) path and reflection path, which run parallel to each other [31], [32]. Figure 2a illustrates the two paths where the satellite and the target are on opposite sides of the receiver. In this figure, r_{LoS} denotes the LoS path length, r_R the reflection path length, and θ represents the satellite elevation. The complex signals from the LoS and reflection paths can be expressed as:

$$S_{LoS} = Ae^{j(\frac{2\pi}{\lambda}r_{LoS} + \varphi_0)}, \quad (2)$$

$$S_R = \beta Ae^{j(\frac{2\pi}{\lambda}r_R + \varphi_0 + \pi)}, \quad (3)$$

where A is the signal amplitude, λ is the carrier wavelength, φ_0 is the initial phase, and $\beta < 1$ is the reflection coefficient, determined by the size and material of the target. This coefficient can be considered constant for a given target. The π in Equation 3 accounts for the phase inversion due to reflection.

As illustrated in Figure 2b, the received signal is a superposition of both signals:

$$S = S_{LoS} + S_R = A'e^{j(\frac{2\pi}{\lambda}r_{LoS} + \tan^{-1}(\frac{\beta \sin \Delta\varphi}{1 + \beta \cos \Delta\varphi}) + \varphi_0)}, \quad (4)$$

where A' is the amplitude of the composite signal. The exponent of e represents the phase, and we denote it as Φ . In commercial GNSS receivers, the signal amplitude is reported as the C/N0, typically rounded to an integer, thereby constraining high-precision sensing capabilities. However, accurate phase Φ can be obtained from the carrier phase measurement reported by the receiver.

We can further decompose Φ into three terms:

$$\begin{aligned} \Phi &= \frac{2\pi}{\lambda}r_{LoS} + \tan^{-1}\left(\frac{\beta \sin \Delta\varphi}{1 + \beta \cos \Delta\varphi}\right) + \varphi_0 \\ &= \frac{2\pi}{\lambda}r_{LoS} + \varphi_M + \varphi_0. \end{aligned} \quad (5)$$

In this equation, $\varphi_M \triangleq \tan^{-1}\left(\frac{\beta \sin \Delta\varphi}{1 + \beta \cos \Delta\varphi}\right)$, where:

$$\begin{aligned} \Delta\varphi &= \frac{2\pi}{\lambda}(r_R - r_{LoS}) + \pi = \frac{2\pi}{\lambda}|P^*P_T P_{Rx}| + \pi \\ &= \frac{4\pi d \cos \theta}{\lambda} + \pi. \end{aligned} \quad (6)$$

P^* , P_T and P_{Rx} are the points shown in Figure 2a. $\Delta\varphi$ is related to the length of the target's reflected path r_R . As φ_M is closely related to the distance d between the target and the receiver, we call φ_M the *target motion signal*.

During subtle target motions, such as respiration, variations in d can cause short-term signal variations. As a person exhales, the chest falls, increasing d from d_1 to d_2 , which results in an increase in the carrier phase to Φ_e . Conversely, during inhalation, the phase shifts to Φ_i , as illustrated in Figure 2c. The correlation between chest motion and carrier phase variation can be utilized for monitoring respiration.

The above analysis is based on the assumption that no satellite movement exists, i.e. the LoS path length remains constant over time in Equation 6. However, in the actual received signal, the variation of LoS path length introduces significant interference in Φ , complicating the detection of subtle target motion embedded in the *target motion signal* φ_M .

Next, We model the interference of satellite movement component in sensing signals. We start by computing the derivative of the phase measurement Φ to examine the temporal characteristics of each dynamic component. According to Equation 6, the derivative of Φ can be expressed as:

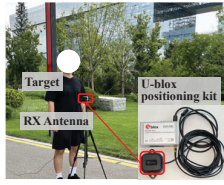
$$\frac{d\Phi}{dt} = \frac{2\pi dr_{LoS}}{\lambda dt} + \frac{d\varphi_M}{dt}, \quad (7)$$

where the first term $\frac{dr_{LoS}}{dt}$ is a large value due to the high-speed of satellite movement, with velocities reaching up to 4 km/s. The second term, $\frac{d\varphi_M}{dt}$, is determined by variations in satellite elevation θ and target-receiver distance d , as delineated in Equation 6. As illustrated in Figure 2d, although the satellite exhibits a high linear velocity, its elevation changes gradually due to its low angular velocity of approximately merely 1.5×10^{-4} rad/s. Consequently, the negligible variation in θ does not contribute to the temporal dynamic of φ_M . Moreover, target motion results in considerably smaller variations

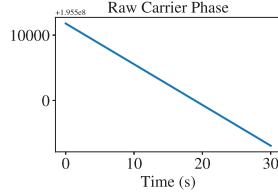
in d than those induced in r_{LoS} by satellite movement. As a result, the second term is much smaller than the first term, i.e.:

$$\frac{d\varphi_M}{dt} \ll \frac{2\pi dr_{LoS}}{\lambda dt}. \quad (8)$$

Figure 2e illustrates the effect of dynamic components on GNSS sensing signals. The satellite movement induces a rapid rotation of the LoS signal, S_{LoS} , characterized by a high angular velocity of $\frac{2\pi dr_{LoS}(t)}{\lambda dt}$. In contrast, the target motion results in only a minor change in the angle, $\Delta\varphi$. Consequently, the *target motion signal* φ_M within Φ is overshadowed by the satellite movement.



(a) Experiment setup.



(b) Raw carrier phase signal.

Fig. 3: Experiment setup and signal with satellite movement.

Experimental verification: We conduct an experiment to demonstrate this phenomenon. As shown in Figure 3a, an U-blox F9 series GNSS receiver is positioned with a subject breathing next to its antenna. According to the analysis in Section III-A, a breathing pattern similar to that in Figure 2c is expected. However, as depicted in Figure 3b, the actual collected signal exhibits a pronounced downward trend due to satellite movement, which overshadows the breathing pattern.

B. Modeling the Interference of Signal Errors on Sensing

GNSS signals suffer from three sources of errors, including satellite-related errors, receiver-related errors, and atmospheric errors. Although existing models address these errors to enhance positioning accuracy [18], they are inadequate for improving sensing performance due to the different impacts of errors on positioning and sensing. Consequently, it is necessary to reconsider these errors within the context of target motion sensing.

Consider satellites p and q transmitting signals, while receivers i and j , located within a relatively short distance (within several kilometers), simultaneously receive signals, as is shown in Figure 4. After accounting for the three sources of errors, the carrier phase received by receiver i from satellite p can be expressed as follows (the propagation path related to satellite p and receiver j follows a similar analysis):

$$\frac{\lambda^p \Phi_i^p}{2\pi} = \frac{\lambda^p \varphi_{M,i}^p}{2\pi} + r_i^p + \delta t^p + A_i^p + \delta t_i + \varepsilon_i^p, \quad (9)$$

where λ^p is the signal wavelength, $\varphi_{M,i}^p$ is the *sensing signal*, r_i^p is the LoS path length, and the remained are signal errors.

As shown in Figure 4, δt^p represents the clock bias of satellite p , which is consistent between receivers i and j . A_i^p accounts for ionospheric and tropospheric errors related to both satellite p and receiver i . δt_i denotes the clock bias of

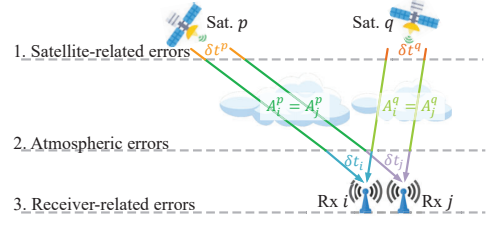


Fig. 4: Effect of different signal errors.

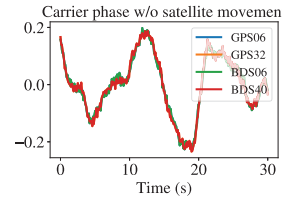
receiver i , which is consistent between satellites p and q . ε_i^p represents the random noise.

The sensing signal $\varphi_{M,i}^p$ on the right hand side is mixed with multiple GNSS errors. The characteristics of these errors are as follows:

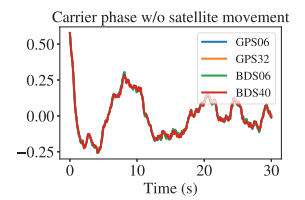
(1) Satellite-related errors (i.e., δt^p and δt^q in Fig. 4) : These errors primarily result from the drift of the satellite's onboard atomic clocks. Within the short-term observation window for target motion, the clock error remains nearly constant due to the stability of atomic clocks, *making its interference on sensing signals insignificant*.

(2) Atmospheric errors (i.e., A_i^p , A_j^p , A_i^q and A_j^q in Fig. 4) : GNSS signals introduce atmospheric errors as they pass through the Earth's atmosphere. These errors stem from variations in atmospheric conditions, such as changes in the ionosphere and troposphere influenced by weather and solar activity. These variations evolve slowly over short periods thereby *have minimal interference on sensing signals*.

(3) Receiver-related errors (i.e., δt_i and δt_j in Fig. 4) : These errors primarily arise from receiver clock biases, which change rapidly due to their simpler electronic design compared to the precise atomic clocks on satellites. These errors significantly interfere with sensing signals, hindering the extraction of target motions. Therefore, *cancelling these errors is essential for high-precision sensing of highly dynamic target motions*. As illustrated in Figure 4, these receiver-related errors are consistent across satellites on the same device but vary between different devices. This provides us with insight into cancelling such errors in Section IV-C.



(a) Signals of device 1.



(b) Signals of device 2.

Fig. 5: Comparison of signals in two receivers under errors.

Experimental verification: The experimental setup is the same as in Section III-A, except two receivers are placed 10 cm apart at the same location for comparison of signals. We cancel the interference of satellite movement (see Section IV-B for details) to focus on signal error characteristics. Figure 5 shows that even after removing satellite motion effects, we

cannot extract the target motion (respiration) pattern from the *carrier phase w/o satellite movement* due to severe errors. Each sub-figure in Figure 5 illustrates that severe signal errors remain constant across four different satellites, irrespective of whether they are part of the Beidou or GPS systems, on the same device. In contrast, a comparison of the two sub-figures reveals that these severe signal errors differ between device 1 and device 2. This suggests that these severe signal errors originate from receiver-related sources, which are the primary obstacle to target motion sensing.

IV. FINE SAT DESIGN

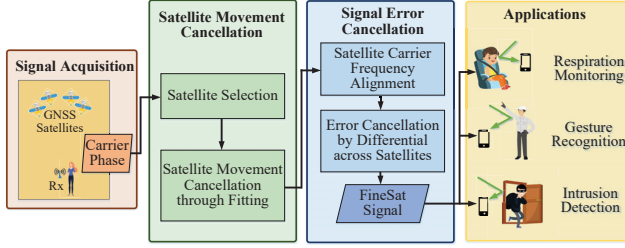


Fig. 6: System overview.

In this section, we present a signal enhancement system named *FineSat* to refine GNSS satellite signals for high-precision target motion sensing. The overview of our system is in Figure 6. *FineSat* comprises three modules: signal acquisition, satellite movement cancellation, and signal error cancellation. Utilizing the enhanced signal, three applications are proposed as demonstrations of its efficacy. In this section, respiration monitoring is employed as an example to illustrate each processing step. The experimental setup mirrors that in Section III-A, and the data flow is depicted in Figure 7.

A. Signal Acquisition

We utilize a commercial positioning device to collect raw signals—C/N0, pseudorange, and carrier phase—from multiple GNSS satellites, following the widely adopted RTCM protocol [29]. The raw data, in binary format, are parsed using the *pyrtcm* package [33] in Python. Specifically, RTCM message types 1077 (GPS data) and 1127 (BeiDou data) are extracted to obtain the carrier phase Φ for further analysis.

B. Satellite Movement Cancellation

Next, we describe our methodology for cancelling satellite movement using polynomial fitting.

Satellite selection. The receiver obtains signals from multiple satellites in various directions. However, in obstructed environments such as urban areas with buildings or trees shown in Figure 3a, signals from certain satellites may be partially blocked, resulting in packet loss and affecting continuous sensing. To address this, we filter out signals with packet loss based on the carrier phase sample rate from the last 30 seconds. Satellites for which the actual sampling rate matches the predetermined rate are reserved, while others are excluded. Figure 7a shows the selection outcomes. Satellites colored in gray have experienced packet loss and are therefore masked,

whereas pink dots denote the reserved satellites. Despite the challenges posed by the urban canyon environment, there are still sufficient satellite signals available for effective sensing thanks to the dense distribution of GNSS satellites.

Satellite movement cancellation through fitting. To mitigate the influence of satellite movement on sensing signals, we propose a novel polynomial fitting approach. The basic idea is that within the temporal window of sensing—such as during a few respiratory cycles—a polynomial function can approximate the satellite’s orbit. A tenth-order polynomial achieves millimeter-level accuracy [34], effectively fitting the satellite’s movement. Meanwhile, in the received signal, the target motion ① exhibits non-polynomial behavior, making it incompatible with polynomial fitting; and ② is far subtler than the satellite movement, as analyzed in Section III-A, thereby does not interfere with the polynomial fitting. As a result, *the fitting residuals* capture the target motion while excluding satellite movement.

Based on this understanding, we present our approach to cancel satellite movement. We define the tenth-order polynomial to be solved as $\hat{r}_{LoS}(t)$, and fit its parameters γ using the least squares algorithm:

$$\arg \min_{\gamma} ||\hat{r}_{LoS}(t) - \frac{\lambda}{2\pi} \Phi(t)|| \quad (10)$$

By fitting Φ , we can approximate r_{LoS} , and the target motion term φ_M stays in the fitting residuals. As shown in Figure 7b, we subtract the fitted satellite movement term $\hat{r}_{LoS}(t)$ from $\Phi(t)$ to acquire fitting residuals, generating *carrier phase signal w/o satellite movement*:

$$\hat{\varphi}_M(t) = \Phi(t) - \frac{2\pi}{\lambda} \hat{r}_{LoS}(t). \quad (11)$$

After this step, the influence of satellite movement is canceled, but GNSS signal errors persist. The respiration patterns are not yet clear in signals, as shown in Figure 7c.

C. Signal Error Cancellation

Then we outline methods for cancelling GNSS errors via signal differential across satellites.

Satellite carrier frequency alignment. This procedure is essential before differential because different satellites broadcast at different frequencies; for instance, the GPS L1 band is at 1575.42 MHz, while the BeiDou B1I band is at 1561.098 MHz. In Equation 9, the errors on the right side are related to the wavelength λ on the left. By matching satellite IDs to their signal frequencies, we calculate the wavelength for each satellite and incorporate it into Equation 9 for alignment.

Error cancellation by differential across satellites. In high-precision positioning, methods like Real-Time Kinematic (RTK) [18] enhance accuracy to the centimeter level by addressing *signal propagation errors*. Two closely located receivers observing the same satellite experience similar propagation errors, which can be eliminated through signal differencing. However, in sensing, it is *receiver-related errors*—such as clock biases and hardware-induced phase noise—that primarily interfere with sensing signals. Since differencing cannot cancel these errors, traditional RTK methods are unsuitable for enhancing sensing signals.

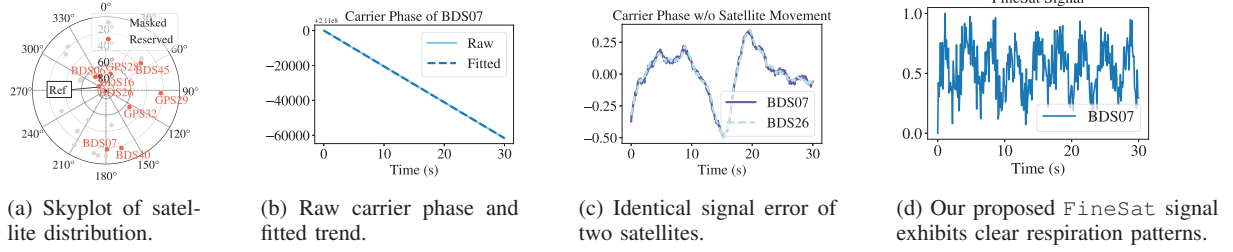


Fig. 7: Data acquisition and processing of FineSat.

To address major errors in sensing signals, we leverage the consistency of receiver-related errors across different satellites observed by the same receiver, as shown in Figure 5. This allows us to use differential techniques across satellites. Specifically, we analyze signals from two satellites, p and q , observed simultaneously by the same receiver i . The *carrier phase w/o satellite movement* of these satellites can be expressed as:

$$\hat{\varphi}_{M,i}^p(t) = \varphi_{M,i}^p(t) + \frac{2\pi}{\lambda^p}(\delta t^p + A_i^p + \delta t_i + \varepsilon_i^p), \quad (12)$$

$$\hat{\varphi}_{M,i}^q(t) = \varphi_{M,i}^q(t) + \frac{2\pi}{\lambda^q}(\delta t^q + A_i^q + \delta t_i + \varepsilon_i^q), \quad (13)$$

where $\varphi_{M,i}^p(t)$ and $\varphi_{M,i}^q(t)$ are the *target motion signals* of two satellites on receiver i , λ^p and λ^q are the wavelengths of two satellites respectively, δt^p and δt^q are the satellite clock biases, A_i^p and A_i^q are atmospheric delays, and δt_i is the clock bias of receiver i , ε_i^p and ε_i^q are random noises.

By subtracting these two signals, the common receiver clock bias δt_i could be cancelled. The **resultant signal $F(t)$, denoted as the FineSat Signal, is utilized for high-precision sensing in our proposed system pipeline.** Disregarding the random noise, it can be written as:

$$\begin{aligned} F(t) &\triangleq \frac{\lambda^p}{2\pi} \hat{\varphi}_{M,i}^p(t) - \frac{\lambda^q}{2\pi} \hat{\varphi}_{M,i}^q(t) \\ &= \frac{\lambda^p}{2\pi} \varphi_{M,i}^p - \left[\frac{\lambda^p}{2\pi} \varphi_{M,i}^q + (\delta t^p - \delta t^q) \right. \\ &\quad \left. + (A_i^p - A_i^q) \right]. \end{aligned} \quad (14)$$

Our dual satellite differential technique cancels the disruptive receiver clock bias while preserving target motion signal of satellite p , i.e. $\frac{\lambda^p}{2\pi} \varphi_{M,i}^p$. Other terms include $\frac{\lambda^p}{2\pi} \varphi_{M,i}^q$ of satellite q , the differential satellite clock bias $(\delta t^p - \delta t^q)$, and the differential atmospheric delay $(A_i^p - A_i^q)$. We select satellite with the highest elevation as satellite q because these satellites are less affected by unwanted multi-path effects [35], thereby reducing potential interference from $\frac{\lambda^p}{2\pi} \varphi_{M,i}^q$ on satellite p 's target motion signal $\frac{\lambda^p}{2\pi} \varphi_{M,i}^p$. Although satellite clock biases and atmospheric errors persist in the FineSat signal, their impact on sensing is can be ignored, as discussed in Section III-B. Satellite clock biases δt^p and δt^q change slowly, remaining nearly constant over the sensing time windows. Likewise, atmospheric errors A_i^p and A_i^q exhibit gradual variations that do not significantly affect the FineSat signal.

The recovered FineSat signal, shown in Figure 7d, clearly reveals nine respiration cycles. With the influences of satellite movement and receiver-related errors cancelled, the FineSat

signal enables the detection of subtle target motions, such as respiration. This enhancement lays the groundwork for various GNSS high-precision sensing applications, which we illustrate through examples including respiration detection, gesture recognition, and intrusion detection.

V. IMPLEMENTATION

We implement the sensing system using a U-blox F9 series positioning device as the GNSS receiver. This device incorporates a widely-used chip found in many consumer electronics [36], facilitating the extension of our system to other consumer devices. We configure the receiver to capture signals from both the GPS and BeiDou systems, ensuring the diversity of satellite signals. The receiver connects to a ThinkPad laptop via USB 2.0 for power and data transmission. The laptop, equipped with an Intel i7 CPU and 16GB of RAM, runs U-blox's u-center software to capture GNSS raw measurements. We set the baud rate to 9,600 and configure the receiver to sample at 10 Hz by default. Signal processing and system implementation are performed using Python 3.9.

VI. EVALUATION

In this section, we assess the performance of FineSat. We begin by examining the computational latency of the signal enhancement process to showcase its real-time processing capabilities. Subsequently, we demonstrate the effectiveness and versatility through three real-world sensing applications: i) outdoor respiration monitoring, ii) robust gesture recognition, and iii) intrusion detection.

A. Computational Latency

We now evaluate the computational latency. Our experiment is conducted using five data samples, each with ten 30-second time windows. Within each window, signals of every reserved satellite undergoes polynomial fitting to cancel satellite movement. Subsequently, the differences between signals of all satellites and the reference satellite are computed. The overall latency is illustrated in Figure 8. The average, medium and maximum latencies are 0.00235, 0.00236 and 0.00332 seconds, respectively, indicating that our method supports real-time processing.

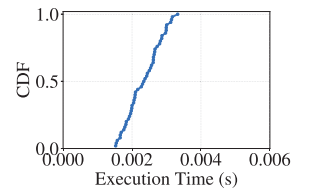


Fig. 8: Overall Latency.

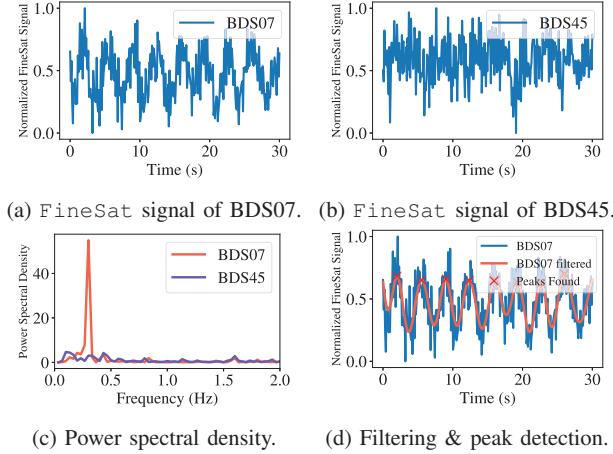


Fig. 9: Data overflow of respiration rate calculation.

B. Case Study 1: Respiration Monitoring

Respiration monitoring is crucial for health assessment. Although Wi-Fi-based systems are effective in indoor environments [37]–[40], they are inadequate in outdoor areas without Wi-Fi coverage. In these scenarios, GNSS signals can serve as a valuable supplementary. For instance, vehicles equipped with GNSS receivers can monitor drivers' respiration rates on roads lacking Wi-Fi access. Likewise, GNSS signals can be employed during activities such as hiking and skiing in locations devoid of other sensing signals. Consequently, we select respiration monitoring as Case Study 1 to demonstrate the effectiveness of our GNSS-based sensing system.

1) *Steps of Respiration Rate Calculation:* We first calculate the respiration rate based on the *FineSat* sensing signal. The specific steps are as follows:

Respiration signal selection: Satellite signal quality varies due to elevation angles and the user's position and orientation. Figures 9a and 9b show two example data sets. Signals from BeiDou07 exhibit clear breathing patterns, while BeiDou45 does not. Therefore, we design an algorithm to select high-quality signals from all available satellites for sensing. Specifically, we use FFT with a window size of 30 seconds to calculate the power spectral density (PSD) of all satellite signals and select the one with the highest peak within the human breathing frequency range (10 to 37 bpm [41]). As shown in Figure 9c, the PSD of BeiDou07 has a significantly higher peak, so we choose this signal for further extraction of the breathing pattern.

Noise reduction: After selecting signals, we apply a high-pass filter with a cutoff frequency of $\frac{1}{15}$ Hz and a moving average filter with a window width of 0.5 seconds to the time-domain signal to reduce noise. As shown in Figure 9d, the blue line represents the raw respiration signal, while the pink line represents the filtered respiration signal.

Respiration rate detection: Subsequently, we calculate the respiration rate by determining the time intervals between the peaks of the respiration signal, as indicated by the red crosses in Figure 9d. Assuming that n peaks are extracted from the

signal over a period of time, let t_i represent the time point of the i -th peak. The estimated respiration rate can then be calculated as: $\frac{60(n-1)}{\sum_{i=1}^{n-1} (t_{i+1} - t_i)}$, with the unit being breaths per minute (bpm).

2) *Experiment Setup:* We test the performance of the respiration monitoring system in four different scenarios: (A) a rural area without Wi-Fi signal coverage, (B) outside an office building, (C) outside another office building, and (D) outside an apartment building, as shown in Figure 10a. The antenna of the U-blox F9 positioning kit is positioned approximately 50 cm from the target. The experiment recruits 8 volunteers, including 3 females and 5 males, with heights ranging from 160 cm to 182 cm. We test the system across different environments (locations A-D), at different times of the day (8:00, 12:00, 16:00 and 20:00 hr.), under various weather conditions (sunny, cloudy, drizzling), and with/without nearby human interference. Each data sample is collected over approximately 60 seconds, and we repeat each experimental set five times. NeuLog NUL-236 respiration monitor belt is used to calculate the ground-truth respiration rate. The mean absolute error (MAE) of the respiration rate serves as the metric for evaluating system's performance.

3) *Overall Performance:* Figure 10b presents the overall performance. Our system achieves an average MAE of 0.42 bpm, being comparable to the 0.47 bpm error observed in Wi-Fi sensing [41]. Additionally, 90% of the data points exhibit an error of less than 0.76 bpm, highlighting the system's high precision in respiration monitoring.

4) *Environment Variations:* The experiment includes four scenarios: the first is a rural environment without Wi-Fi coverage, and the other three are urban canyons where buildings and trees may obstruct satellite signals. Environments B, C, and D are located under an 18-story office building, a 15-story office building, and in a residential area surrounded by 15-story buildings, respectively. For location A shown in Figure 10a, Figure 10c shows our system achieves an average MAE of 0.36 bpm, demonstrating the GNSS sensing system's pervasiveness. In environments B, C, and D, the average absolute errors are 0.33, 0.39, and 0.41 bpm, respectively. Despite potential obstructions, *FineSat* consistently measures respiration accurately by leveraging the broad distribution of GNSS satellites, which enables the selection of unobstructed signals for monitoring.

5) *User Variations:* To validate the robustness of *FineSat* to user diversity, we invite three females and five males to participate as volunteers in testing the respiration monitoring system. The experimental results are shown in Figure 10d. The MAE of respiration for all eight volunteers is below 0.38 bpm. This demonstrates that our *FineSat* system is robust to variations among subjects.

6) *Impact of Satellite Distribution:* The distribution of GNSS satellites in space varies at different times of the day. To validate the robustness of *FineSat* under different GNSS satellite distributions, we test the effectiveness of respiration monitoring at various times (approximately 8:00, 12:00, 16:00, and 20:00 hr.). The experimental results are shown in Fig-

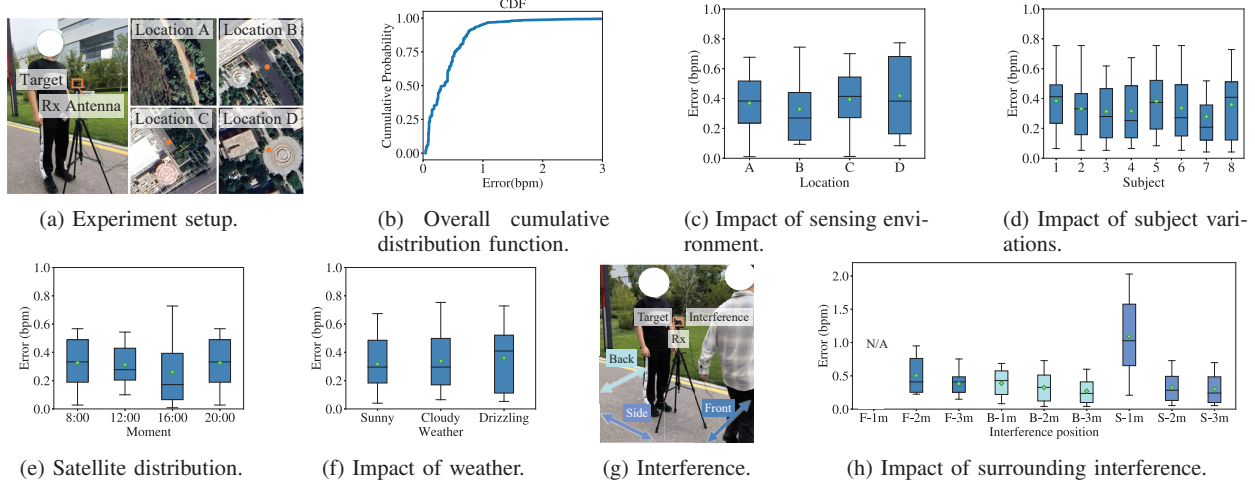


Fig. 10: Experiment setups and results of Case 1: Respiration Detection.

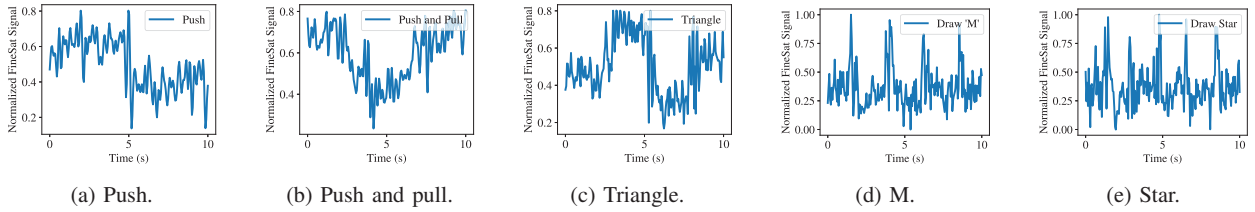


Fig. 11: Examples of FineSat signals for five different gestures.

ure 10e, with the MAE at these times all below 0.33 bpm. This result shows that despite the varying satellite distributions, it is always possible to select high-quality respiration signals from the constellations for precise respiration monitoring.

7) *Impact of Weather Conditions*: The propagation of GNSS signals is affected differently by the atmosphere under varying weather conditions. To investigate whether this factor impacts high-precision sensing, we conducted tests under three weather conditions: sunny, cloudy, and drizzling. The results are shown in Figure 10f. The mean absolute errors were 0.32, 0.36, and 0.34 bpm, respectively. These results indicate that despite the atmospheric differences, the system consistently maintained good performance. This finding aligns with our understanding of atmospheric errors discussed in Section III-B, where we concluded that such errors do not significantly interfere with signals in the short term.

8) *Impact of Surrounding Interference*: To evaluate our system's robustness against interference, we position the interferer at various distances and positions relative to the target, as shown in Figure 10g. Tests are conducted with the interferer in front (F), beside (S), and behind (B) the target at 1, 2, and 3 meters from the antenna. Figure 10h presents the results. The system performs worst with the interferer in front at 1 meter, failing to detect respiration. This happens because the system relies on signals reflected from the target's chest, and obstruction by the interferer degrades performance. Conversely, interference is less severe when the interferer is beside or behind the target. Additionally, greater distances between the interferer and target reduce interference

and measurement error. For instance, with the interferer at the side at 1, 2, and 3 meters, errors are 1.09 bpm, 0.33 bpm, and 0.30 bpm, respectively. Thus, the system maintains high detection accuracy in most scenarios.

C. Case Study 2: Gesture Recognition

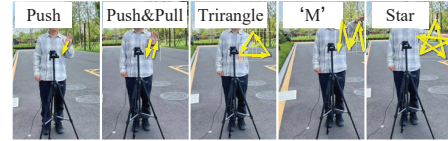
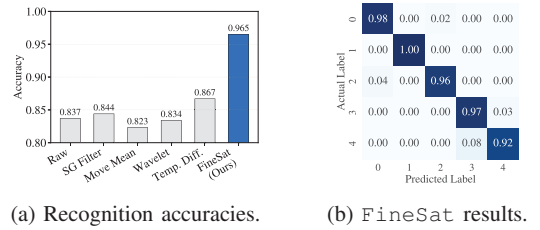


Fig. 12: Illustration of five gestures.



(a) Recognition accuracies. (b) FineSat results.

Fig. 13: Experiment results of gesture recognition.

Gesture recognition represents an important approach in human-computer interaction [38], [39], [42], [43]. Leveraging GNSS signals for gesture recognition can broaden its applicability to a wider range of scenarios that lack other sensing signals, without incurring the costs of additional sensing equipment. For example, in-vehicle GNSS receivers can be repurposed to control the trunk or manage audio systems. In

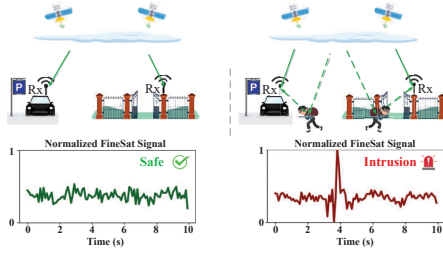


Fig. 14: Intrusion detection with GNSS signals.

Case Study 2, we demonstrate the effectiveness of utilizing FineSat signals for gesture recognition.

We study five common gestures: Push, Push&Pull, Triangle, M, and Star, as shown in Figure 12. After processing with FineSat, the signals corresponding to the five gestures are shown in Figure 11. We utilize Support Vector Machine (SVM) [44], a widely studied machine learning model, for our gesture recognition task. We collect data at different times and locations, involving five volunteers and resulting in a total of 1000 training samples, with 200 samples for each gesture. We allocate 80% of the data for training and 20% for testing, and perform five-fold cross-validation.

To demonstrate the effectiveness of FineSat in enhancing GNSS sensing signals, we benchmark our approach against both raw signals and advanced signal processing techniques utilized in state-of-the-art sensing systems [39]. The baseline signal processing techniques include the Savitzky-Golay Filter [45], Moving Mean Filter [46], Wavelet Transform [47], and Temporal Difference [46]. The results are shown in Figure 13. As depicted in Figure 13a, the accuracy of the model trained with the raw GNSS carrier phase is only 83.7%. Furthermore, the baseline signal processing techniques fail to significantly enhance the accuracy, with results ranging from 82.3% to 86.7%. In contrast, the model trained with FineSat signals achieves an accuracy of 96.5%. This improvement is attributed to FineSat's ability to mitigate signal fluctuations due to receiver-related GNSS errors, thereby enabling the classifier to accurately identify variations caused by different gestures, which is a capability not matched by any of the baseline methods.

D. Case Study 3: Intrusion Detection

Intrusion detection is crucial for maintaining security across various scenarios, including home and vehicle protection. For instance, car manufacturers are incorporating Sentry Mode to guard security during parking [48]. However, the current Sentry Mode, which relies on cameras, raises concerns regarding power consumption [48]. Additionally, it often compromises user privacy and struggles in low-light conditions [49]. We aim to address these limitations by utilizing GNSS signals for intrusion detection. This approach leverages existing GNSS equipments in homes and vehicles, allowing costless integration into current systems. In Case Study 3, we illustrate the implementation of a GNSS-based intrusion detection system, as shown in Figure 14.

We collect a GNSS signal-based intrusion detection dataset from six participants with heights ranging from 160 cm to 181 cm, comprising a total of 2,200 data entries, of which 576 are intrusion events and 1,624 are non-intrusion events. Our dataset covers four different environments and three types of weather conditions (sunny, cloudy, and drizzling). Four different environments are shown in Figure 15. We allocate 80% of the data for training and 20% for testing, and perform five-fold cross-validation.

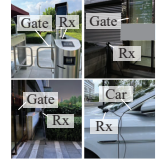
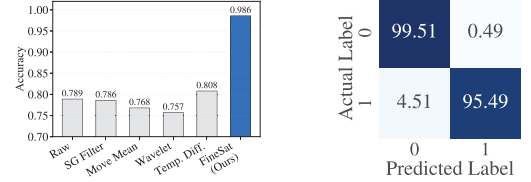


Fig. 15: Setups.



(a) Detection accuracies.

(b) FineSat results.

Fig. 16: Experiment results of intrusion detection.

We also utilize SVM [44] as the classifier, training and testing with both the raw carrier phase and the FineSat signal. The experimental results are shown in Figure 16. Figure 16a presents the accuracies of the baseline methods mentioned in Section VI-C alongside our approach. The accuracy with raw GNSS carrier phase signals is merely 78.9%, and it ranges from 75.7% to 80.8% across all baseline methods. Such low accuracies hinder reliable intrusion detection. In contrast, with data processed by FineSat, the accuracy increases significantly to 98.6%. As shown in the confusion matrix in Figure 16b (label=1: intrusion; label=0: no intrusion), the False Positive Rate is 0.49%, while the True Positive Rate reaches 95.49%. This improvement is attributed to FineSat's ability to eliminate device clock errors in raw signals, enabling the classifier to detect signal variations caused by intrusions.

VII. DISCUSSION

A. Issue of Sensing Distance

The long propagation distance of signals, combined with energy attenuation during reflection from targets, results in weak GNSS received signals. This can limit the effective sensing distance of our system, particularly since it relies on commodity GNSS receivers with relatively simple antennas. However, this limitation has a dual aspect: the reduced sensing distance also makes the system less susceptible to surrounding interference. To extend the sensing distance, several strategies can be employed, including the strategic placement of receiver antennas [50], leveraging the diversity of sensing signals [51], utilizing beamforming techniques to enhance the SNR [52], and implementing advanced antenna designs [53].

B. Adapting GNSS Sensing for Indoor Environments

FineSat leverages ubiquitous GNSS signals to monitor fine-grained respiration and coarse-grained intrusion. How-

ever, indoor signal blockage may limit its performance. Recent research has explored predicting indoor GNSS signal strength using the 3D layout of buildings [54]. This helps identify suitable GNSS sensing areas indoors. In regions where GNSS signals are weak, alternative strategies, such as using repeaters to relay signals [55], highlight the potential to extend *FineSat*'s capabilities to indoor applications.

C. Extending GNSS Sensing to Other Satellite Systems

FineSat is designed to leverage satellite signals within the GNSS, including GPS, BeiDou and Galileo. Notably, our results show negligible differences in performance when using signals from different navigation systems, demonstrating the versatility and robustness of the system. In addition to GNSS satellites, various other satellites, like communication satellites and meteorological satellites, are continuously transmitting signals to Earth. Their signal structures have already been studied for positioning purposes [56]. After adjusting accordingly to their characteristics, we believe our techniques can also be extended to these satellites.

VIII. RELATED WORK

A. GNSS Remote Sensing

Beyond navigation and positioning [18], [19], [25], [57], the extensive coverage of GNSS signals has inspired researchers to explore their properties for remote sensing. In 1993, Manuel first proposed using GPS reflected signals for remote sensing [58]. Subsequent research leveraged the superposition of multi-path signals to measure parameters such as sea level, snow thickness and sea ice thickness [20]–[22]. By analyzing signal amplitude, phase and polarization with dedicated devices, researchers have also monitored ocean wind speed [59] and soil moisture [23]. Additionally, studies have explored the use of GNSS signals received by airborne and spaceborne receivers, further expanding the scope of GNSS remote sensing [20], [60]. Unlike these studies focusing on static environmental objects, we aim to enable GNSS signals for sensing highly dynamic human motions with high precision.

B. Wireless Sensing

In recent years, wireless signals such as Wi-Fi [9], [46], [61], LTE [14], LoRa [62], RFID [15], and various radar waveforms [1], [7], [8], [63]–[66] have been widely applied in various sensing tasks. Among them, Wi-Fi is particularly appealing due to its widespread deployment, with access points already extensively available. Wi-Fi signals have been leveraged for a variety of sensing tasks, such as respiration monitoring [41], heart rate detection [67], tracking [8], and human activity recognition [42]. However, its sensing coverage is restricted to areas around Wi-Fi access points, thereby limiting its pervasiveness [68]. In contrast, LoRa signals provide a broader coverage range around its access points than Wi-Fi [62], [69]. Nonetheless, LoRa also encounters limitations in areas lacking access point existence. Compared to these signals, GNSS signals offer global coverage without the need for deploying access points, thereby exhibiting exceptional pervasiveness. Recently, Dong et al. [32] utilized

downloaded ephemeris data, signal strength, and carrier phase to reconstruct GNSS signals for sensing. In contrast, our work systematically models GNSS interferences for target motion sensing based on satellite orbit and signal features. Building on this model, we employ innovative polynomial fitting and cross-satellite differentials to mitigate the interferences. Our approach eliminates the reliance on downloaded ephemeris data, tackling the challenges in areas lacking internet access to that data. Meanwhile, GNSS signal strength is no longer required in this work, making *FineSat* more suitable for integration into commercial protocols that only provide coarse, integer-level signal strength [28], [29].

C. Differential GNSS

Differential GNSS is a well-established technique for enhancing positioning and tracking accuracy [70]. Its fundamental principle involves using one or more reference receivers to calculate GNSS error corrections [71]. Real-time kinematic (RTK) positioning, a specific type of differential GNSS, provides real-time carrier phase error corrections, achieving centimeter-level positioning accuracy [72]. Recently, Hedgecock et al. [73] utilized a network of smartphones to implement differential GNSS for precise tracking, eliminating the need for dedicated reference receivers. Besides, Gowda et al. [74] proposed a novel GNSS-based method for drone orientation tracking by applying double differencing across receivers and satellites, as well as across receivers and time. In contrast to these studies, which primarily focus on enhancing positioning and tracking accuracy, our research emphasizes the application of differential techniques to address GNSS interference in high-precision sensing. Our approach broadens the scope of differential GNSS applications, extending its utility beyond positioning accuracy improvements.

IX. CONCLUSION

In conclusion, this study presents *FineSat*, a GNSS signal enhancement system that leverages commercial devices for high-precision target motion sensing. We introduce a comprehensive GNSS sensing model to analyze sensing interference of satellite movement and three primary sources of errors: satellite-related, receiver-related, and atmospheric. Based on this model, we develop a polynomial fitting module in our system to mitigate satellite movement interference and a signal differential module to address severe receiver-related errors. Our system's effectiveness is validated through extensive experiments across diverse subjects and environments, achieving a mean absolute error of 0.42 bpm in respiration monitoring, 96.5% accuracy in gesture recognition, and 98.6% accuracy in intrusion detection, demonstrating its effectiveness for high-precision sensing applications.

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REFERENCES

- [1] Z. Juncen, J. Cao, Y. Yang, W. Ren, and H. Han, "mmdrive: Fine-grained fatigue driving detection using mmwave radar," *ACM Transactions on Internet of Things*, vol. 4, no. 4, pp. 1–30, 2023.
- [2] D. Zhang, X. Zhang, S. Li, Y. Xie, Y. Li, X. Wang, and D. Zhang, "Lt-fall: The design and implementation of a life-threatening fall detection and alarming system," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 7, no. 1, pp. 1–24, 2023.
- [3] X. Zhang, D. Zhang, Y. Xie, D. Wu, Y. Li, and D. Zhang, "Waffle: A waterproof mmwave-based human sensing system inside bathrooms with running water," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 7, no. 4, pp. 1–29, 2024.
- [4] S. Cao, D. Li, S. I. Lee, and J. Xiong, "Powerphone: Unleashing the acoustic sensing capability of smartphones," in *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*, 2023, pp. 1–16.
- [5] W. Chen, C. Zheng, W. He, Y. Zou, and K. Wu, "Ultra write: A lightweight continuous gesture input system with ultrasonic signals on cots devices," in *2024 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. IEEE, 2024, pp. 174–183.
- [6] K.-J. Butkow, T. Dang, A. Ferlini, D. Ma, and C. Mascolo, "heart: Motion-resilient heart rate monitoring with in-ear microphones," in *2023 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. IEEE, 2023, pp. 200–209.
- [7] Y. Cao, A. Dhekne, and M. Ammar, "Utrack3d: 3d tracking using ultra-wideband (uwb) radios," in *Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services*, 2024, pp. 345–358.
- [8] A. Nomura, M. Sugasaki, K. Tsubouchi, N. Nishio, and M. Shimomura, "Device-free multi-person indoor localization using the change of tof," in *2023 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. IEEE, 2023, pp. 190–199.
- [9] F. Adib and D. Katabi, "See through walls with wifi!" in *Proceedings of the ACM SIGCOMM 2013 conference on SIGCOMM*, 2013, pp. 75–86.
- [10] H. Wang, D. Zhang, J. Ma, Y. Wang, Y. Wang, D. Wu, T. Gu, and B. Xie, "Human respiration detection with commodity wifi devices: Do user location and body orientation matter?" in *Proceedings of the 2016 ACM international joint conference on pervasive and ubiquitous computing*, 2016, pp. 25–36.
- [11] X. Li, S. Li, D. Zhang, J. Xiong, Y. Wang, and H. Mei, "Dynamic-music: Accurate device-free indoor localization," in *Proceedings of the 2016 ACM international joint conference on pervasive and ubiquitous computing*, 2016, pp. 196–207.
- [12] X. Li, D. Zhang, Q. Lv, J. Xiong, S. Li, Y. Zhang, and H. Mei, "Indotrack: Device-free indoor human tracking with commodity wi-fi," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 1, no. 3, pp. 1–22, 2017.
- [13] M. Cominelli, F. Gringoli, and F. Restuccia, "Exposing the csi: A systematic investigation of csi-based wi-fi sensing capabilities and limitations," in *2023 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. IEEE, 2023, pp. 81–90.
- [14] W. Chen, K. Niu, D. Zhao, R. Zheng, D. Wu, W. Wang, L. Wang, and D. Zhang, "Robust dynamic hand gesture interaction using lte terminals," in *2020 19th ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN)*. IEEE, 2020, pp. 109–120.
- [15] C. Duan, J. Liu, X. Ding, Z. Li, and Y. Liu, "Full-dimension relative positioning for rfid-enabled self-checkout services," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 5, no. 1, pp. 1–23, 2021.
- [16] Z. El Khaled, H. Mcchick, and F. Petrillo, "Wifi coverage range characterization for smart space applications," in *2019 IEEE/ACM 1st International Workshop on Software Engineering Research & Practices for the Internet of Things (SERP4IoT)*. IEEE, 2019, pp. 61–68.
- [17] A. Ghosh, J. Zhang, J. G. Andrews, and R. Muhamed, *Fundamentals of LTE*. Pearson Education, 2010.
- [18] A. Leick, *GPS satellite surveying*. Wiley, 2015.
- [19] B. Hofmann-Wellenhof, H. Lichtenegger, and E. Wasle, *GNSS—global navigation satellite systems: GPS, GLONASS, Galileo, and more*. Springer Science & Business Media, 2007.
- [20] H. Qiu and S. Jin, "Global mean sea surface height estimated from spaceborne cyclone-gnss reflectometry," *Remote Sensing*, vol. 12, no. 3, p. 356, 2020.
- [21] S. Yang, S. Jin, Y. Jia, and M. Ye, "Significant wave height estimation from joint cygnss ddma and les observations," *Sensors*, vol. 21, no. 18, p. 6123, 2021.
- [22] A. Alonso-Arroyo, V. U. Zavorotny, and A. Camps, "Sea ice detection using uk tds-1 gnss-r data," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 55, no. 9, pp. 4989–5001, 2017.
- [23] X. Wu, Z. Dong, S. Jin, Y. He, Y. Song, W. Ma, and L. Yang, "First measurement of soil freeze/thaw cycles in the tibetan plateau using cygnss gnss-r data," *Remote Sensing*, vol. 12, no. 15, p. 2361, 2020.
- [24] S. C. Bhardwaj, S. Shekhar, A. Vidyarthi, and R. Prakash, "Satellite navigation and sources of errors in positioning: a review," in *2020 International Conference on Advances in Computing, Communication & Materials (ICACCM)*. IEEE, 2020, pp. 43–50.
- [25] G. Xu and Y. Xu, *GPS*. Springer, 2007.
- [26] A. N. Idris, A. M. Suldi, J. R. A. Hamid, and D. Sathyamoorthy, "Effect of radio frequency interference (rfi) on the global positioning system (gps) signals," in *2013 IEEE 9th international colloquium on signal processing and its applications*. IEEE, 2013, pp. 199–204.
- [27] G. X. Gao and P. Enge, "How many gnss satellites are too many?" *IEEE Transactions on aerospace and electronic Systems*, vol. 48, no. 4, pp. 2865–2874, 2012.
- [28] R. Langley, "Nmea 0183: A gps receiver," *GPS world*, vol. 6, no. 7, pp. 54–57, 1995.
- [29] The Radio Technical Commission for Maritime Services, "Rtcm published standards & publications," <https://www.rtcmm.org/publications>.
- [30] L. Weill, "C/a code pseudorange accuracy-how good can it get?" in *Proceedings of the 7th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GPS)*, 1994, pp. 133–141.
- [31] N. Roussel, F. Frappart, G. Ramillien, J. Darrozes, C. Desjardins, P. Gegout, F. Pérosanz, and R. Biancale, "Simulations of direct and reflected wave trajectories for ground-based gnss-r experiments," *Geoscientific model development*, vol. 7, no. 5, pp. 2261–2279, 2014.
- [32] H. Dong, M. Cui, N. Wang, L. Qiu, J. Xiong, and W. Wang, "Gpsense: Passive sensing with pervasive gps signals," in *Proceedings of the 30th Annual International Conference on Mobile Computing And Networking*, 2024, pp. 1000–1014.
- [33] "Rtcm3 protocol parser," <https://pypi.org/project/pyrtcm/>.
- [34] M. Horemuz and J. V. Andersson, "Polynomial interpolation of gps satellite coordinates," *GPS solutions*, vol. 10, pp. 67–72, 2006.
- [35] K. Zheng, X. Zhang, P. Li, X. Li, M. Ge, F. Guo, J. Sang, and H. Schuh, "Multipath extraction and mitigation for high-rate multi-gnss precise point positioning," *Journal of Geodesy*, vol. 93, pp. 2037–2051, 2019.
- [36] u-blox, "Annual report 2023," <https://content.u-blox.com/sites/default/files/documents/Annual-Report-2023.pdf>.
- [37] Y. Zeng, D. Wu, R. Gao, T. Gu, and D. Zhang, "Fullbreathe: Full human respiration detection exploiting complementarity of csi phase and amplitude of wifi signals," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 2, no. 3, pp. 1–19, 2018.
- [38] J. Liu, H. Liu, Y. Chen, Y. Wang, and C. Wang, "Wireless sensing for human activity: A survey," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 3, pp. 1629–1645, 2020.
- [39] S. M. Hernandez and E. Bulut, "Wifi sensing on the edge: Signal processing techniques and challenges for real-world systems," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 1, pp. 46–76, 2022.
- [40] X. Wang, A. Yu, K. Niu, W. Shi, J. Wang, Z. Yao, R. C. Shah, H. Lu, and D. Zhang, "Understanding the diffraction model in static multipath-rich environments for wifi sensing system design," *IEEE Transactions on Mobile Computing*, vol. 23, no. 11, pp. 10393–10410, 2024.
- [41] Y. Zeng, D. Wu, J. Xiong, E. Yi, R. Gao, and D. Zhang, "Farsense: Pushing the range limit of wifi-based respiration sensing with csi ratio of two antennas," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 3, no. 3, pp. 1–26, 2019.
- [42] Y. Liu, A. Yu, L. Wang, B. Guo, Y. Li, E. Yi, and D. Zhang, "Unifi: A unified framework for generalizable gesture recognition with wi-fi signals using consistency-guided multi-view networks," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 7, no. 4, pp. 1–29, 2024.
- [43] Z. Hong, Z. Li, S. Zhong, W. Lyu, H. Wang, Y. Ding, T. He, and D. Zhang, "Crosshar: Generalizing cross-dataset human activity recognition via hierarchical self-supervised pretraining," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 8, no. 2, pp. 1–26, 2024.

- [44] S. Suthaharan and S. Suthaharan, "Support vector machine," *Machine learning models and algorithms for big data classification: thinking with examples for effective learning*, pp. 207–235, 2016.
- [45] X. Wang, K. Niu, A. Yu, J. Xiong, Z. Yao, J. Wang, W. Li, and D. Zhang, "Wimeasure: Millimeter-level object size measurement with commodity wifi devices," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 7, no. 2, pp. 1–26, 2023.
- [46] W. Li, R. Gao, J. Xiong, J. Zhou, L. Wang, X. Mao, E. Yi, and D. Zhang, "Wifi-csi difference paradigm: Achieving efficient doppler speed estimation for passive tracking," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 8, no. 2, pp. 1–29, 2024.
- [47] X. Wang, C. Yang, and S. Mao, "On csi-based vital sign monitoring using commodity wifi," *ACM Transactions on Computing for Healthcare*, vol. 1, no. 3, pp. 1–27, 2020.
- [48] I. Tesla, "Sentery mode, model 3 owner's manual," 2023. [Online]. Available: https://www.tesla.com/ownersmanual/model3/en_us/GUID-56703182-8191-4DAE-AF07-2FDC0EB64663.html
- [49] M. Abdalla, S. Javed, M. A. Radi, A. Ulhaq, and N. Werghi, "Video anomaly detection in 10 years: A survey and outlook," 2024. [Online]. Available: <https://arxiv.org/abs/2405.19387>
- [50] X. Wang, K. Niu, J. Xiong, B. Qian, Z. Yao, T. Lou, and D. Zhang, "Placement matters: Understanding the effects of device placement for wifi sensing," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 6, no. 1, pp. 1–25, 2022.
- [51] Y. Li, D. Wu, J. Zhang, X. Xu, Y. Xie, T. Gu, and D. Zhang, "Diversense: Maximizing wi-fi sensing range leveraging signal diversity," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 6, no. 2, pp. 1–28, 2022.
- [52] M. H. Keshvadi, A. Broumandan, and G. Lachapelle, "Analysis of gnss beamforming and angle of arrival estimation in multipath environments," in *Proceedings of the 2011 International Technical Meeting of The Institute of Navigation*, 2011, pp. 427–435.
- [53] J. J. Wang, "Antennas for global navigation satellite system (gnss)," *Proceedings of the IEEE*, vol. 100, no. 7, pp. 2349–2355, 2012.
- [54] H. Zhou and T. Maekawa, "Predicting signal reception information from gnss satellites in indoor environments without site survey: Towards opportunistic indoor positioning based on gnss fingerprinting," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 8, no. 3, pp. 1–30, 2024.
- [55] H. Dong, Y. Xie, X. Zhang, W. Wang, X. Zhang, and J. He, "Gpsmirror: Expanding accurate gps positioning to shadowed and indoor regions with backscatter," in *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*, 2023, pp. 1–15.
- [56] T. E. Humphreys, P. A. Iannucci, Z. M. Komodromos, and A. M. Graff, "Signal structure of the starlink ku-band downlink," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 5, pp. 6016–6030, 2023.
- [57] Z. Hong, G. Wang, W. Lyu, B. Guo, Y. Ding, H. Wang, S. Wang, Y. Liu, and D. Zhang, "Cominer: nationwide behavior-driven unsupervised spatial coordinate mining from uncertain delivery events," in *Proceedings of the 30th International Conference on Advances in Geographic Information Systems*, 2022, pp. 1–10.
- [58] M. Martin-Neira *et al.*, "A passive reflectometry and interferometry system (paris): Application to ocean altimetry," *ESA journal*, vol. 17, no. 4, pp. 331–355, 1993.
- [59] V. U. Zavorotny and A. G. Voronovich, "Scattering of gps signals from the ocean with wind remote sensing application," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 38, no. 2, pp. 951–964, 2000.
- [60] N. Rodriguez-Alvarez, D. M. Akos, V. U. Zavorotny, J. A. Smith, A. Camps, and C. W. Fairall, "Airborne gnss-r wind retrievals using delay-doppler maps," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 51, no. 1, pp. 626–641, 2013.
- [61] Z. Yao, X. Wang, K. Niu, R. Zheng, J. Wang, and D. Zhang, "Wiprofile: Unlocking diffraction effects for sub-centimeter target profiling using commodity wifi devices," in *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking*, 2024, pp. 185–199.
- [62] H.-C. Lee and K.-H. Ke, "Monitoring of large-area iot sensors using a lora wireless mesh network system: Design and evaluation," *IEEE Transactions on Instrumentation and Measurement*, vol. 67, no. 9, pp. 2177–2187, 2018.
- [63] Z. Chang, F. Zhang, X. Ma, P. Wang, W. Chen, D. Zhang, B. Jouaber, and D. Zhang, "Mmecare: Enabling fine-grained vital sign monitoring for emergency care with handheld mmwave radars," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 8, no. 4, pp. 1–24, 2024.
- [64] P. Wang, X. Ma, R. Zheng, L. Chen, X. Zhang, D. Zeghlache, and D. Zhang, "Slprof: Improving the temporal coverage and robustness of rf-based vital sign monitoring during sleep," *IEEE Transactions on Mobile Computing*, 2023.
- [65] X. Ma, P. Wang, J. Liu, and D. Zhang, "Micro vibration reconstruction under in-range large-scale dynamic clutters using a bi-exponential radar signal model," in *2024 IEEE/MTT-S International Microwave Symposium-IMS 2024*. IEEE, 2024, pp. 386–389.
- [66] L. Lu, X. Ma, Y. Liang, Z. Liu, X. Fan, and L. Li, "A 60-ghz hybrid fmcw-doppler radar for vibration detection with a robust i/q calibration method," *IEEE Sensors Journal*, vol. 22, no. 21, pp. 20 464–20 474, 2022.
- [67] A. Khamis, C. T. Chou, B. Kusy, and W. Hu, "Cardiofi: Enabling heart rate monitoring on unmodified cots wifi devices," in *Proceedings of the 15th EAI international conference on mobile and ubiquitous systems: Computing, networking and services*, 2018, pp. 97–106.
- [68] M. F. Khan, G. Wang, M. Z. A. Bhuiyan, and X. Li, "Wi-fi signal coverage distance estimation in collapsed structures," in *2017 IEEE International Symposium on Parallel and Distributed Processing with Applications and 2017 IEEE International Conference on Ubiquitous Computing and Communications (ISPA/IUCC)*. IEEE, 2017, pp. 1066–1073.
- [69] F. Zhang, Z. Chang, K. Niu, J. Xiong, B. Jin, Q. Lv, and D. Zhang, "Exploring lora for long-range through-wall sensing," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 4, no. 2, pp. 1–27, 2020.
- [70] P. Teunissen and A. Khodabandeh, "Review and principles of ppp-rtk methods," *Journal of Geodesy*, vol. 89, no. 3, pp. 217–240, 2015.
- [71] G. Morgan-Owen and G. Johnston, "Differential gps positioning," *Electronics & Communication Engineering Journal*, vol. 7, no. 1, pp. 11–21, 1995.
- [72] X. Li, J. Huang, X. Li, Z. Shen, J. Han, L. Li, and B. Wang, "Review of ppp-rtk: Achievements, challenges, and opportunities," *Satellite navigation*, vol. 3, no. 1, p. 28, 2022.
- [73] W. Hedgecock, M. Maroti, A. Ledeczi, P. Volgyesi, and R. Banalagay, "Accurate real-time relative localization using single-frequency gps," in *Proceedings of the 12th ACM Conference on Embedded Network Sensor Systems*, 2014, pp. 206–220.
- [74] M. Gowda, J. Manweiler, A. Dhekne, R. R. Choudhury, and J. D. Weisz, "Tracking drone orientation with multiple gps receivers," in *Proceedings of the 22nd annual international conference on mobile computing and networking*, 2016, pp. 280–293.